

What a Language Model Does with a Solver Certificate: An Answer-Evidence Decomposition Across Twelve Models

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Abstract

Solver-, tool- and retrieval-augmented systems report large accuracy gains, routinely read as evidence that the model used the supplied material. Two mechanisms give the same number: reading a supplied answer, and reasoning over supplied state. The standard control, a same-shape irrelevant record, separates neither. We build one that does, holding every surface property we could enumerate and audit fixed while destroying validity alone, across four sealed experiments and two robustness checks: 34,584 scored responses, re-derivable from released receipts.

Asked what a model does with a certificate, the instrument returns a different answer for almost every checkpoint, arm and inference runtime. The share of the state effect that survives destroying validity is 98.3% on one checkpoint and 40.4–51.4% on another. Inference depth is one step on two checkpoints and not on a third; one model is a usable interface under one inference stack and falls below the viability floor under another, on identical weights. Two criteria we had prespecified were gamed by label collapse.

One result does not move. Given a certificate whose fabricated final rule establishes the query’s negation, all three checkpoints answer incorrectly on 189, 186 and 192 of 192 items, and the strongest abstains on 0 of 192 broken certificates. Mechanism claims about solver assistance do not transfer between checkpoints. The failure to check the solver does.

1 Introduction

Pairing a language model with an external solver, tool or retrieval index reliably raises accuracy on tasks the model cannot do alone. Systems that translate a problem into a formal representation, dispatch it to a symbolic engine and return the verdict through the model report large end-to-end gains [Pan et al., 2023, Olausson et al., 2023, Ye et al., 2023, Gao et al., 2022, Schick et al., 2023, Yang et al., 2023, Lewis et al., 2020]. Those gains are routinely read as evidence that the model *used* the supplied material.

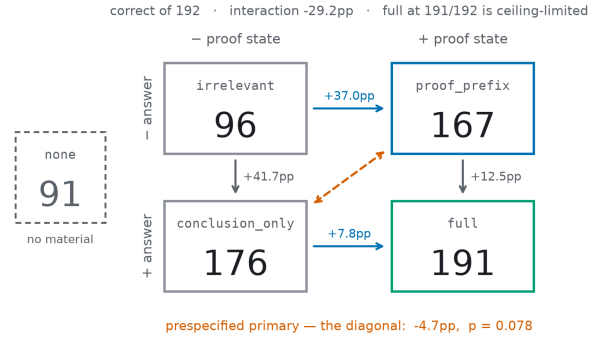


Figure 1: The instrument. A solver certificate carries an answer and the state it follows from, and accuracy adds the two together. Arranging the arms as a 2×2 , across by whether the record carries the answer and down by whether it carries the state, makes every arrow a one-factor edge that attributes. Cells give correct items of 192 for the reanalysed experiment. The dashed diagonal moves both factors at once, which is why its null attributed to neither.

That reading is underdetermined. Under *answer reading* the supplied record contains the conclusion and the model copies it out; under *state use* the model works with the supplied premises and completes the inference itself. Both give the same accuracy. The standard control, a same-shape but irrelevant record, rules out proof length, formatting and generic validity, but not answer reading: a prefix stopping one step short still names the query’s subject and predicate in adjacent lines, so a model that only checks co-occurrence reproduces every number a reasoning model would.

The distinction is not academic. A model completing a supplied inference has some purchase on whether that inference is sound; a model matching surface co-occurrence has none, and anything built on it inherits the solver’s errors in silence. No accuracy number separates the two.

Separating them needs a control that holds surface form fixed and destroys validity alone. We build one and apply it across four sealed experiments and two robustness checks on frozen small models, scoring by direct next-

token likelihood over verified single-token candidates so that no free text is parsed.

The instrument works, and almost nothing it returns transfers. Breaking validity alone removes 98.3% of the state effect on one checkpoint and 40.4–51.4% on another. Inference depth, order-sensitivity and whether a model can serve as the interface at all move the same way: they are properties of a checkpoint, an arm and an inference runtime, not of the architecture. We therefore offer the method and no general value, ours included. The boundary is strict, too: on the Qwen checkpoints every arm without a near-complete chain returns $d' = 0$, so high accuracy with a full certificate describes a faithful transcriber rather than a reasoner.

One result does not move. Given a certificate whose fabricated final rule establishes the query’s negation, all three checkpoints answer incorrectly on 189, 186 and 192 of 192 items, and none abstains on a broken one. Mechanism claims about solver assistance do not transfer between checkpoints. The failure to check the solver does.

What is new is the instrument: a 2×2 separating solver-supplied answers from solver-supplied state (fig. 1), a surface-matched invalid certificate certified per item, and a closure certifier for queries nothing decides. We apply it to twelve models [Qwen Team et al., 2024, Grattafiori et al., 2024, Gemma Team et al., 2025, Microsoft et al., 2025, OLMo Team et al., 2024, Allal et al., 2025].

2 Related Work

Chain-of-thought made intermediate steps part of the output [Wei et al., 2022], and a literature then asked whether the answer rests on them: explanations omit the biasing features that drive the prediction [Turpin et al., 2023], injected mistakes are often ignored [Lanham et al., 2023, Arcuschin et al., 2025], answer-determining hints go unverballed [Chen et al., 2025], and one response is to make the trace a program a solver executes [Lyu et al., 2023]. Neurosymbolic work puts the formal step outside the model altogether [Pan et al., 2023, Olausson et al., 2023, Ye et al., 2023, Gao et al., 2022], and the pattern extends to tool use [Schick et al., 2023], premise selection [Yang et al., 2023] and retrieval [Lewis et al., 2020].

Both leave our question open. The first intervenes on *model-generated* traces, where ignoring a mistake is unfaithful; ours is externally supplied, formally verified state, where following a corrupted one is unsafe. The second reports end-to-end accuracy, which cannot separate the solver’s *answer* from its *state* and assumes the solver correct, though the model side is not neutral to context [Shi et al., 2023, Xie et al., 2023].

The task family descends from rule-based deductive datasets in natural language [Clark et al., 2020, Tafjord et al., 2020, Saparov and He, 2022, Han et al., 2022]. Order effects there are large [Chen et al., 2024], the sets predate the models evaluated on them [Deng et al., 2024, Golchin and Surdeanu, 2023], and counterfactual variants show how much apparent competence rests on familiarity [Wu et al., 2023]. None supplies the matched adversarial certificates an attribution question needs, so we generate items after model release with per-item invented vocabularies and build the certificates ourselves.

Prompted classification is biased rather than merely noisy: accuracy swings with format and example order [Zhao et al., 2021], option order alone moves it [Pezeshkpour and Hruschka, 2023, Zheng et al., 2023], likelihood scoring lets strings compete for probability mass independently of correctness [Holtzman et al., 2021], and abstention is its own capability [Kadavath et al., 2022, Yin et al., 2023]. On a balanced panel a degenerate single-label responder is indistinguishable from a chance-level competent one, so we report d' and criterion c for every arm [Macmillan and Creelman, 1991, Stanislaw and Todorov, 1999] with the exact McNemar test [McNemar, 1947] and bootstrap intervals [Efron, 1987]. Two cases here motivate that rather than illustrate it: an arm scoring 50.0% with $d' = 0.00$, and a prespecified criterion of our own passed by models answering one label almost everywhere.

3 Method

3.1 The answer-evidence decomposition

A solver certificate carries two things: the *answer* to the query, and the *state* that answer follows from. End-to-end accuracy conflates them. We separate them with a 2×2 of answer evidence by proof state (fig. 1), with a no-material baseline outside it. `IRRELEVANT` supplies a valid derivation for a different entity, `CONCLUSION_ONLY` the derived terminal literal alone, `PROOF_PREFIX` the chain with that literal withheld, and `FULL` both. An *edge* moves one factor and therefore attributes; the *diagonal* moves two and cannot. The experiment we reanalyse chose that diagonal as its primary contrast.

Why the usual control is not enough. Take a theory in which Kexil is dovrant, every dovrant thing is plesh, and every plesh thing is marn, with the query *is Kexil marn?* The standard control gives a same-shape derivation about a different entity, fixing length, format and generic validity. But the arm it is compared against ends on *Kexil is plesh*, one line from a rule naming *marn*: the query’s subject and its predicate sit in adjacent lines. A model that never performs the last step, and only registers that the two

Table 1: What the model sees, from sealed item b15-4ef53c71. Every prompt is the same 18-line theory in invented vocabulary, then one solver record, then the query. Only the record changes between arms. Left: the FULL record, a complete four-step derivation. Right: what each other arm does to it. BROKEN_CHAIN is the word-matching test — it replaces line 6 with a different *real* rule of the theory, one that does not fire, so the chain stops reaching the query, while line 8 still names the query’s predicate and the entity still appears four times, exactly as in TRUNCATE_1. MISLEADING instead invents a rule the theory does not contain.

Query: Bimol is vrinpek. (true)	
FULL record	
1 Bimol is vrasler.	NONE
2 All vrasler people are deinmal.	IRRELEVANT
3 Bimol is deinmal.	SAME_ENTITY_IRREL.
4 All deinmal people are kuthbror.	TRUNCATE_1
5 Bimol is kuthbror.	TRUNCATE_2
6 All kuthbror people are gasplaek.	TRUNCATE_3
7 Bimol is gasplaek.	SHUFFLED
8 All gasplaek people are vrinpek.	BROKEN_CHAIN
9 Bimol is vrinpek.	
	MISLEADING
	FULL
	Record supplied
	no record at all
	same shape, a different entity throughout
	same entity, an unrelated predicate
	lines 1–8: the answer line withheld
	lines 1–6: two steps withheld
	lines 1–4: three steps withheld
	lines 1–8 in seeded random order
	lines 1–8, but 6 becomes All foskdem people are zuftbas. and 7 becomes Bimol is zuftbas. Line 8 unchanged
	lines 1–9, but 8 becomes All gasplaek people are not vrinpek. and 9 becomes Bimol is not vrinpek.
	lines 1–9

terms co-occur where a derivation is heading, reproduces every number a model completing it would. The square localises the gain to state; it cannot show the state was used as inference.

3.2 Surface-matched invalidity: the word-matching test

Table 1 shows one sealed item and every record we supply for it. Call the prefix that stops one step short TRUNCATE_1: it is the strongest arm that withholds the answer, and the one a surface-matching model can still exploit. BROKEN_CHAIN is built to strip that exploit. We cut one intermediate link, so the displayed lines no longer reach the query, while holding fixed against TRUNCATE_1 the query-entity occurrence count (audited per item), the query predicate in the final rule, the line count and the fact/rule/literal sequence. Every rule displayed is a real rule of the theory; exactly one derived literal is false. All 192 items are certified to leave the query undecided under the displayed lines. That gives

$$\text{total state} = \text{TRUNCATE_1} - \text{IRRELEVANT}, \quad (1)$$

$$\text{surface} = \text{BROKEN_CHAIN} - \text{IRRELEVANT}, \quad (2)$$

$$\text{validity} = \text{TRUNCATE_1} - \text{BROKEN_CHAIN}. \quad (3)$$

Five companion arms separate the rest. SAME_ENTITY_IRRELEVANT isolates entity repetition, naming the query subject four times where IRRELEVANT names it none. SHUFFLED keeps TRUNCATE_1’s eight lines in seeded random order. TRUNCATE_2 and TRUNCATE_3 withhold two and three steps, giving a depth ladder. And MISLEADING is the corrupted-solver arm: one fabricated final rule, absent from the theory, establishes

the query’s negation, so following the supplied record means answering incorrectly while the true answer is unchanged.

3.3 Certifying that nothing decides a query

Showing a query is derivable is easy: exhibit the proof. Showing nothing decides it requires saturating the theory’s forward closure. Our certifier works under open-world semantics with explicit negation, so a negative literal holds only if something derives it, never by failure to prove the positive. It agrees with a public corpus [Tafford et al., 2020] on 23,240 of 23,240 questions, including 10,440 undecided; passes nine adversarial theories probing cycles, round-cap overruns, derived negation and absent symbols; and agrees with an independently written reference on 192 of 192 items. No item enters a sealed panel without a certification record.

4 Experimental Setup

Every experiment asks a rule-chaining query over a small theory of facts and universally quantified rules. Table 2 names the six and says what each one does.

The abstention rescoring generates no panel. It rescores the validity control’s items with a third candidate available, changing only the instruction line and the candidate set on byte-identical inputs, and its NONE arm fixes the abstention floor under that same instruction. On the sweep’s 64 undecidable items there is no terminal literal to supply, so those two cells instantiate the 2×2 less exactly than the rest. Two robustness checks reuse these panels: one rescores the validity control’s items under eight permutations of

Table 2: The four experiments and two robustness checks. Sealed panels use per-item invented vocabularies, are mutually disjoint, and fix derivation depth at four. The public corpus validates the certifier only: no item from it is scored by any model here.

Panel	Items	Experiment	What it does
b14	192 binary	the reanalysis	re-reads a published five-arm factorial as the 2×2 of fig. 1
b15	192 binary	the validity control	ten matched arms (table 1) on three checkpoints
b16	192 three-class	the model sweep	ten models $\times$ five arms, with an <code>Unknown</code> class
b17, b17b	b15 items	the abstention rescoring	re-scores the broken certificates with <code>Unknown</code> allowed
b18	b15 items	line-order check	eight permutations of each record
b16 on CUDA	as b16	runtime check	<code>transformers</code> rather than <code>mlx-lm</code> , plus two larger models
ProofWriter OWA	23,240	certifier validation only [Tafjord et al., 2020]	

the certificate’s lines each, recording where the final rule and its premise land; the other rescores the sweep’s panel through `transformers` on CUDA rather than `mlx-lm`, and there adds Llama-3.1-8B and Qwen2.5-14B.

Responses are scored by direct next-token likelihood over verified single-token candidates: `Yes/No` for the first two, `Yes/No/Unknown` for the others. The model generates no free text, so nothing is parsed and no verdict is inferred from prose. Every prompt is instruction, theory, solver record and query, and the theory is visible in every arm. In the reanalysis the record’s header differs between arms as well as its content, a confound on that experiment’s primary contrast, removed everywhere after it.

Models. The reanalysis and the validity control use one frozen 3B instruction-tuned checkpoint at 4-bit precision [Qwen Team et al., 2024], pinned to byte-identical weights. The sweep scores ten instruction-tuned models from 0.5B to 7.6B across six families, Qwen2.5, Llama-3.2, Gemma-3, Phi-4-mini, OLMo-2 and SmoLLM2 [Qwen Team et al., 2024, Grattafiori et al., 2024, Gemma Team et al., 2025, Microsoft et al., 2025, OLMo Team et al., 2024, Allal et al., 2025], through `mlx-lm` from pinned mirrors; the runner did not record a dtype, so we claim no uniform precision across the ten. No gate blocks that run: delivery is itself the capability under test, so a model that cannot emit a verdict reliably is a finding rather than an exclusion.

Sensitivity, not accuracy. Accuracy on a balanced panel is uninterpretable when the responder is biased. One arm below scores 50.0% with an identical 9/96 “Yes” rate in *both* classes: $d' = 0.00$, no discriminative signal at all, at or below answering one label to everything. We therefore report d' and criterion c for every arm [Macmillan and Creelman, 1991, Stanislaw and Todorov, 1999] using the loglinear correction, balanced accuracy for the three-class panel (chance 33.3%), the exact two-sided McNemar test [McNemar, 1947] as the decision criterion, and seeded paired bootstrap intervals [Efron, 1987] adjusted for multiple comparisons within declared families.

5 Results

5.1 The state effect is real, and it is validity

The reanalysis begins from a null. Its prespecified primary contrast, `PROOF_PREFIX` – `CONCLUSION_ONLY` = -4.7pp (exact McNemar $p = 0.078$), was reported as no effect. Read through the square, that contrast is the diagonal: it moves both factors at once, so the null attributes to neither.

Both channels carry real effect, and they are not additive. The state main effect is $+22.4\text{pp}$, the answer main effect $+27.1\text{pp}$, and the interaction -29.2pp , larger than either; table 3 gives the four edges. This reading is post hoc, re-derived from the 960 raw receipts by a script that asserts it reproduces all five published per-arm counts before reporting. The design is ceiling-limited, `FULL` reaching 191/192, so both “other factor present” edges are compressed and the averaged main effects deflated with them.

The baseline description was also wrong. The model answers `No` to 88% of items, and `IRRELEVANT` returns an identical 9/96 “Yes” rate in both classes at $d' = 0.00$ (table 4), below a trivial always-`No` strategy’s 96/192.

Destroying validity alone removes almost all of the state effect. `TRUNCATE_1` – `BROKEN_CHAIN` = $+60.4\text{pp}$ (exact McNemar $p = 6.9 \times 10^{-18}$, BCa $[+49.0, +68.8]$, entailed subset $n = 96$ as declared before the run), with d' dropping 2.446, so this is a change in discrimination and not in criterion. The total state effect of $+61.5\text{pp}$ is therefore $+1.0\text{pp}$ of surface overlap ($p = 1$) and $+60.4\text{pp}$ of validity: a **validity share of 98.3%**, 95% percentile $[94.4, 100.0]$. Table 5 gives the full arm set on three checkpoints.

Two supporting contrasts. Entity repetition explains nothing ($+0.0\text{pp}$, $p = 1$) despite one arm naming the query subject four times and the other not at all. And depth is a cliff rather than a slope on the Qwen checkpoints: withholding two steps instead of one scores identically to supplying no certificate, 0/96. Gemma-3-4B is not in that position, and the direction there is worse than null: a two-step-truncated certificate leaves it at 4/96 against 22/96 unaided.

Table 3: The four one-factor edges of the 2×2 , and the prespecified diagonal. Post hoc reanalysis of sealed data.

Contrast	Effect	Exact McNemar p	Reported originally
Add answer, no state present	+41.67pp	1.7×10^{-24}	no
Add state, no answer present	+36.98pp	1.5×10^{-19}	secondary
Add answer, state present	+12.50pp	1.2×10^{-7}	no
Add state, answer present	+7.81pp	6.1×10^{-5}	secondary
<i>Diagonal (prespecified primary)</i>	−4.69pp	0.078	primary

Table 4: Signal detection for the reanalysis; hits and false alarms out of 96 each. Apparent chance and zero sensitivity are not the same thing.

Arm	Hits	FA	d'	c
NONE	9	14	−0.25	+1.17
IRRELEVANT	9	9	0.00	+1.29
CONCLUSION_ONLY	80	0	+3.52	+0.81
PROOF_PREFIX	71	0	+3.20	+0.97
FULL	95	0	+4.72	+0.20

5.2 Almost nothing the instrument returns transfers

The instrument transfers; the number it returns does not. The decomposition was run on two further checkpoints: the same model at bfloat16, which isolates quantisation from model identity, and a different family. Figure 2 plots the split.

Quantisation is the smaller effect and leaves the share unchanged. Model identity is not like that. Gemma-3-4B answers `BROKEN_CHAIN` correctly on 58 of 96 items where the other two manage 1, so the same control that isolates inference on one checkpoint exposes substantial surface exploitation on another. Its share is 40.4% against the `IRRELEVANT` control and 51.4% against what it manages unaided; the divergence is the finding rather than either number. The control is not inert on a model like that either, costing Gemma most of its own competence, so any effect measured against it there is inflated. Order-sensitivity does not replicate alongside the share either (table 5).

The square itself is no steadier. Read across ten models, post hoc from receipts already collected, the mean state main effect is +8.7pp against a mean answer main effect of +21.6pp: the answer channel dominates, and the reanalysis’s +22.4pp is not typical of the set. Excluding four responders degenerate in every arm gives +14.7pp against +28.5pp, and the ordering survives either subset.

Whether a model can serve as the interface at all varies the same way. We had declared that a model relays non-determination at $\geq 80\%$ recall on undecidable items with full solver material. Eight of ten passed and the criterion was invalid: single-class recall is maximised by answering

Unknown to everything, which four models approach. We replace it **post hoc** with a floor on every class, minimum per-class recall ≥ 0.50 , retaining both verdicts in the released results. A second prespecified criterion, a paired `FULL` — `NONE` of 20pp on the undecidable items, is met by five of ten under Holm, two of which the replacement floor rejects: the same gameability by another route. Table 6 gives all five arms for all twelve models.

Read by row, viability is a property of (model $\times$ arm $\times$ runtime), not of a model. Qwen2.5-7B falls from second best in the sweep under `CONCLUSION_ONLY` to 0.141 minimum per-class recall under `FULL`: supplying the proof state is what breaks it, and it breaks three of ten models here. Nothing below 3B clears the floor in any arm, but that is a floor for *tolerating full certificates* rather than a size law: Llama-3.1-8B is degenerate in every arm, and Qwen2.5-14B never clears it.

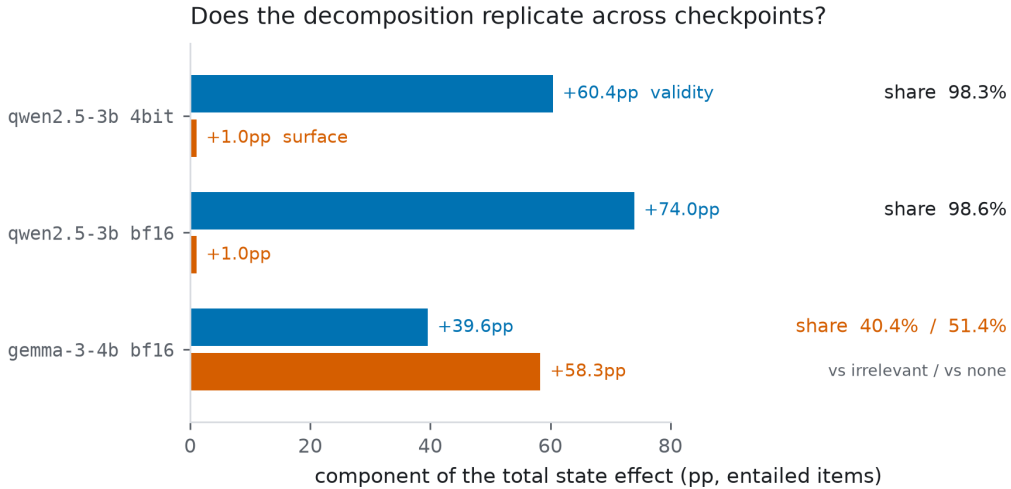
The third factor is the inference stack. Re-scoring the same sealed panel, same weights and same prompt bytes through transformers on CUDA covers nine of the ten. Eight agree on 95.5–99.7% of individual responses with no verdict moving. Phi-4-mini agrees on 87.9%, and its `FULL` arm is viable under one stack and not the other: minimum per-class recall 0.719 against 0.219, on identical weights, with flips reaching 3.0 logits. Gemma-3-4B, the checkpoint carrying the 40.4–51.4% share, is the one model this check cannot cover, so that share has no runtime replication.

5.3 What does not vary

Reversing the final rule and the premise it fires on costs accuracy on every checkpoint. Separating them does not. The split is post hoc: `SHUFFLED` is one seeded permutation per item and moves three things at once, so we re-scored the same items under eight permutations each on all three checkpoints, recording where those two lines landed. An identity arm reproduces `TRUNCATE_1`’s bytes and returns its receipts 192/192 throughout. Distance is flat; direction is not (fig. 3). Rule *after* its premise gives 26.8, 22.4 and 94.9% across the three checkpoints against 5.8, 4.8 and 84.9% for rule *before*; paired within item on the 4-bit checkpoint, 45 items favour rule-after against 2

Table 5: The validity control: ten matched arms scored on three checkpoints, entailed subset (correct of 96, with signed d'). Only the solver record differs between arms. The share rows give the validity component as a fraction of the total state effect, against the IRRELEVANT control and against the unaided NONE arm; the two coincide wherever unaided competence is zero.

Arm	Qwen2.5-3B 4-bit		Qwen2.5-3B bf16		Gemma-3-4B	
	/96	d'	/96	d'	/96	d'
NONE	0	0.00	0	0.00	22	1.83
IRRELEVANT	0	0.00	0	0.00	2	0.62
SAME_ENTITY_IRRELEVANT	0	0.00	0	0.00	3	0.77
TRUNCATE_3	0	0.00	0	0.00	4	0.88
TRUNCATE_2	0	0.00	0	0.00	4	0.88
TRUNCATE_1	59	2.85	72	3.23	96	5.13
BROKEN_CHAIN	1	0.41	1	0.41	58	2.83
SHUFFLED	14	1.53	12	1.43	86	3.80
MISLEADING	0	−4.36	0	−4.06	0	−5.13
FULL	96	5.13	96	5.13	96	5.13
Validity share vs IRRELEVANT	98.3%		98.6%		40.4%	
Validity share vs NONE	98.3%		98.6%		51.4%	



The 98.3% share does not generalise: gemma-3-4b's surface component is +58.3pp against +1.0pp on both qwen checkpoints. Its share is 40.4% against the same-shape control and 51.4% against what it manages unaided — the two denominators diverge only where a substrate has competence of its own.

Figure 2: The surface and validity components on three checkpoints. Gemma-3-4B's surface component is +58.3pp against +1.0pp for the two Qwen checkpoints. Bars give the decomposition against the IRRELEVANT control; the share annotation gives both denominators, 40.4% against that control and 51.4% against what the model manages unaided. The two coincide on the Qwen checkpoints, which score 0/96 with no certificate, and diverge only where a model has competence of its own.

($p = 1.6 \times 10^{-11}$). The direction holds everywhere, its size does not: a $4.6\times$ ratio on the Qwen checkpoints is a 10pp modulation on Gemma-3-4B.

Nothing defends against a wrong solver. The MISLEADING arm is identical to FULL except in its last two sentences: a fabricated final rule, absent from the theory, and the inverted literal it licenses. The ground-truth answer is unchanged. All three checkpoints answer incorrectly on 189, 186 and 192 of 192 items, at d' of −4.36, −4.06

and −5.13, so the model exploiting surface overlap most is also the one most completely misled. The class split is sharper still. This model's standing prior is No: under NONE it answers No to all 192 items, at $c = 2.565$. Under MISLEADING it answers Yes on 93 of the 96 contradicted items. A fabricated rule inverts a near-total prior.

Whether any model notices, we cannot say. Rescoring the BROKEN_CHAIN items with an Unknown candidate available should raise abstention if the model tracks validity. It

Table 6: All five arms, all twelve models. Each cell is balanced accuracy (% , chance = 33.3) / minimum per-class recall; bold clears the 0.50 recall floor, so VIABLE IN is derivable from the cells beside it. Shaded rows are the scale extension on a second runtime: the parameter ladder continues, the numbers are not comparable across the rule. Viability is per arm because it is a property of (model $\times$ arm $\times$ runtime), not of a model.

Model	B	NONE	IRREL	CONCL	PREFIX	FULL	VIABLE IN
Qwen2.5-0.5B	0.5	35.4/0.00	34.9/0.00	41.1/0.00	43.2/0.00	63.5/0.00	—
OLMo-2-1B	1.0	35.9/0.00	33.3/0.00	53.6/0.00	34.9/0.00	42.2/0.00	—
Llama-3.2-1B	1.2	33.9/0.00	33.9/0.00	57.8/0.00	36.5/0.00	37.0/0.00	—
Qwen2.5-1.5B	1.5	33.3/0.00	33.3/0.00	43.2/0.00	34.9/0.00	37.0/0.00	—
SmolLM2-1.7B	1.7	33.3/0.00	33.3/0.00	45.3/0.00	39.6/0.00	69.3/0.38	—
Qwen2.5-3B	3.0	41.1/0.00	39.1/0.00	96.4/0.91	74.0/0.56	90.6/0.72	concl, prefix, full
Llama-3.2-3B	3.2	45.8/0.00	43.2/0.02	67.2/0.02	55.2/0.02	80.7/0.42	—
Phi-4-mini	3.8	44.8/0.00	39.6/0.00	68.2/0.05	71.9/0.33	90.6/0.72	full
Gemma-3-4B	4.3	63.5/0.36	51.0/0.06	100.0/1.00	84.4/0.53	96.4/0.89	concl, prefix, full
Qwen2.5-7B	7.6	33.3/0.00	33.3/0.00	97.9/0.94	67.2/0.02	71.4/0.14	concl only
Llama-3.1-8B	8.0	38.5/0.00	34.4/0.00	65.6/0.00	64.6/0.00	66.7/0.00	—
Qwen2.5-14B	14.7	57.3/0.00	35.4/0.00	80.7/0.42	70.3/0.12	82.8/0.48	—

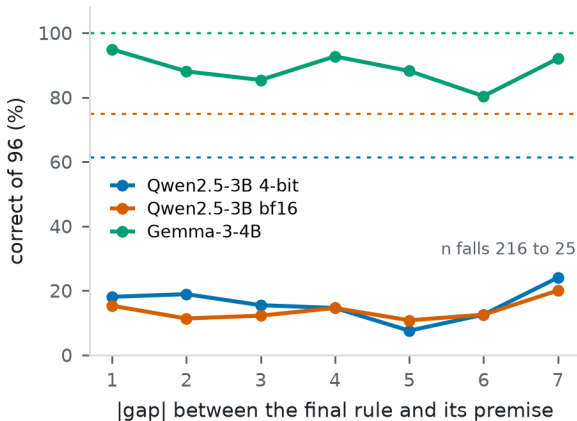


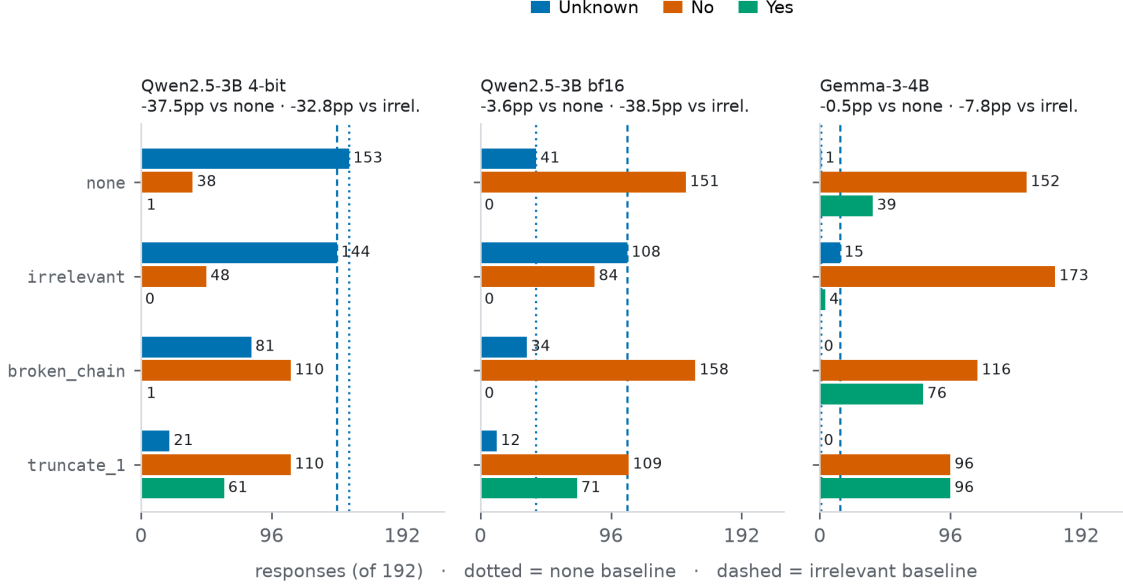
Figure 3: Accuracy against the gap between the final rule and the premise it fires on: all seven gaps, all three checkpoints, each checkpoint’s canonical-order score dashed. Distance is flat. Direction is not: rule-after scores 26.8, 22.4 and 94.9% against 5.8, 4.8 and 84.9% for rule-before. Gemma-3-4B still answers 84.9% of rule-before items correctly, so reversal degrades rather than prevents.

falls by 37.5 and 32.8 points against NONE and IRRELEVANT and rises by 31.2 against the surface-matched TRUNCATE_1 (fig. 4): the sign is a function of the reference, and neither reference is clean. We draw no conclusion from it and withdraw the claim that it refutes detection. What needs no reference is that the most capable of the three abstains 0 times in 192 on a broken certificate and answers Yes on 76. We find no positive evidence that any checkpoint verifies validity, and the direct evidence is the corruption arm above, which a model checking validity would not fail.

6 Limitations

Scope of the mechanism claim. On both Qwen checkpoints the inference is exactly one step deep: withholding two steps scores identically to supplying no certificate (0/96). Gemma-3-4B is not in that position (4/96 against 22/96 unaided), so depth is checkpoint-specific, as is the share, which spans 40.4–98.6% against the IRRELEVANT control and 51.4–98.6% against unaided. The full theory is visible in every arm, so breaking a certificate is diagnostic only where the model scores 0/96 from the theory alone, true of both Qwen checkpoints and not of Gemma-3-4B, where 20 of its 22 unaided successes recur under BROKEN_CHAIN; a model using a broken certificate as a pointer back into the theory would be doing inference of a different kind, and we record that as an interpretation caveat rather than a controlled alternative. TRUNCATE_2 and TRUNCATE_3 also drop the query predicate, so the depth ladder confounds depth with predicate presence, and an undecidable PROOF_PREFIX cannot contain that predicate either, so such items offer fewer surface cues.

Post hoc analyses and superseded criteria. The 2×2 reading of the reanalysis, the sweep’s replacement viability criterion and the gap-versus-direction split are all post hoc and labelled where used; the originals are retained in the released results. The reanalysis carries five defects we report rather than repair: a prespecified 15pp target with only 8.3pp of headroom above the 91.7% arm it applied to; no recorded proof depth, so its failures cannot be stratified; task identifiers encoding the semantic class; a release as a flat receipt export with no sealed panel, so its answer key is reconstructed from those identifiers; and a screen for response tokens leaking into a certificate that was a malformed regular expression and never fired (the released



On every checkpoint, against either reference, a broken chain lowers abstention rather than raising it. Gemma-3-4B abstains on 0 of 192 broken chains and completes 76 of them.

Figure 4: Three-candidate response distribution, three checkpoints, four arms. Dotted and dashed lines mark the *NONE* and *IRRELEVANT* abstention baselines, 4.7pp apart at 4-bit and 34.9pp at bfloat16. A broken chain lowers abstention against *both* references on every checkpoint, the opposite of the direction validity detection predicts.

certificates contain none, checked after the fact). Every later experiment fixes all five.

Generalisation. One task family, small models, synthetic panels. Invented vocabularies establish item novelty, not independence from every relevant pretraining pattern [Golchin and Surdeanu, 2023, Deng et al., 2024], and model is not randomised, so cross-model comparisons are descriptive. The abstention rescaling changes the instruction and candidate set, and response distributions move substantially [Zhao et al., 2021, Zheng et al., 2023], so its contrasts hold but its absolute rates do not. The reanalysis and the validity control use a 4-bit checkpoint and the model sweep bfloat16, so they describe different artefacts of one model, and the sweep found which models could do the job before the validity control’s checkpoints were chosen, which left Phi-4-mini unrun on those arms. Every response is a forced choice among verified single-token candidates, which no deployed solver-augmented setup does. At $n = 96$ the loglinear correction censors d' at ± 5.13 and eight cells of table 5 sit on it, so d' differences involving a saturated arm are bounds. The reanalysis’s per-arm counts and primary contrast appeared in an earlier unpublished report by the present authors, which concluded the 3B checkpoint failed an open-world unknown gate where the sweep finds the family viable; differing panels, candidate sets and quantisation plausibly

explain it, but we flag it rather than leave it to a reader who finds both. Finally, every theory here arrives *already formalised*: nothing tests whether a model can turn a prose framework into solver input, which the corruption result makes consequential.

7 Conclusion

End-to-end accuracy cannot say why solver assistance helps. The decomposition can: arrange the arms so each contrast moves one factor, and break validity while holding surface form fixed. Applied across twelve models, two inference stacks and 34,584 responses, it returns a different answer almost every time it is asked. The validity share is 98.3% on one checkpoint and 40.4–51.4% on another; inference depth, order-sensitivity and viability itself are properties of a checkpoint, an arm and a runtime. We offer the method and decline to generalise its value, ours included.

One result does not move. Given a certificate whose fabricated final rule establishes the negation, the three checkpoints answer incorrectly on 189, 186 and 192 of 192, inheriting the apparatus’s errors in full, and none abstains on a broken certificate. Mechanism claims about solver assistance do not transfer between checkpoints;

the failure to check the solver does. Report both, per checkpoint, and always run a corrupted arm. Report sensitivity beside accuracy while doing it: two criteria we had prespecified were gamed by label collapse, and a baseline arm at 50.0% had $d' = 0.00$.

Reproducibility Statement

Code and data. The supplementary material accompanying this submission is self-contained: sealed panels and answer authorities, all 34,584 per-response receipts, the generators, runners, audits and analysis scripts, and the forward-closure certifier with its validation against the public corpus. Every number in this paper can be re-derived from a fresh unpack with no network access, no model weights and no external packages; the audits use the Python standard library only, so verification does not depend on resolving an environment. Panels are hash-stamped, and the runners for the validity control and the model sweep refuse a panel whose hash does not match, while the others record the hash without enforcing it. The answer authority is opened after all receipts are written, by convention in the analysis order rather than by a gate in the code. Scoring takes the highest-likelihood candidate token; on an exact tie the first candidate in the declared order wins, which across the validity control’s 5,760 prospective responses occurs twice. Each receipt carries a SHA-256 of its own contents, and all 34,584 verified with zero failures. The count is not typed into the manuscript: it is produced by `scripts/census_receipts_v1.py` from the released receipts and substituted at compose time.

Checking one number takes a minute. `scripts/verify.py` recomputes each headline value and prints it beside the value this paper states. Eight of its ten claims re-derive from the receipts and the sealed panels; the cross-model mean and the model table read released analysis files, which the listed scripts regenerate: `python3 scripts/verify.py validity-share` recomputes the 98.3% of section 3.2 and shows the four counts it came from. `--all` adds Gemma’s two baselines, the corruption counts, the detection direction on three checkpoints, the 2×2 state main effect, the cross-model mean, the response census, a recomputed digest for every one of the 34,584 receipts, all sixty cells of table 6, and the ten printed properties of the worked example in table 1. It uses the standard library only, so a reviewer needs no environment beyond Python. Model weights are not redistributed, but each run records the pinned revision and a hash of the weight files before those files are deleted. A permanent archive with a DOI will accompany the camera-ready version.

AI-Use Statement

A large language model was used as a coding and drafting assistant throughout: generating experiment code, analysis scripts and figure code, and drafting this manuscript. All experimental design decisions, the interpretation of results, and the claim boundaries were made and verified by the authors. Every numeric value reported was produced by the released analysis scripts from sealed receipts, not written by a model; the released audits allow any reader to re-derive them independently.

References

- Loubna Ben Allal, Anton Lozhkov, Elie Bakouch, et al. SmoLM2: When smol goes big – data-centric training of a small language model. *arXiv preprint arXiv:2502.02737*, 2025. doi: 10.48550/arxiv.2502.02737.
- Iván Arcuschin, Jett Janiak, Robert Krzyzanowski, Senthoooran Rajamanoharan, Neel Nanda, and Arthur Conmy. Chain-of-thought reasoning in the wild is not always faithful. *arXiv preprint arXiv:2503.08679*, 2025. doi: 10.48550/arxiv.2503.08679.
- Xinyun Chen, Ryan A. Chi, Xuezhi Wang, and Denny Zhou. Premise order matters in reasoning with large language models. *arXiv preprint arXiv:2402.08939*, 2024. doi: 10.48550/arxiv.2402.08939.
- Yanda Chen, Joe Benton, Ansh Radhakrishnan, et al. Reasoning models don’t always say what they think. *arXiv preprint arXiv:2505.05410*, 2025. doi: 10.48550/arxiv.2505.05410.
- Peter Clark, Oyvind Tafjord, and Kyle Richardson. Transformers as soft reasoners over language. *arXiv preprint arXiv:2002.05867*, 2020. doi: 10.24963/ijcai.2020/537.
- Chunyuan Deng, Yilun Zhao, Xiangru Tang, Mark Gestein, and Arman Cohan. Investigating data contamination in modern benchmarks for large language models. *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 2024. doi: 10.18653/v1/2024.naacl-long.482.
- Bradley Efron. Better bootstrap confidence intervals. *Journal of the American Statistical Association*, 1987. doi: 10.1080/01621459.1987.10478410.
- Luyu Gao, Aman Madaan, Shuyan Zhou, et al. Pal: Program-aided language models. *arXiv preprint*

- arXiv:2211.10435*, 2022. doi: 10.48550/arxiv.2211.10435.
- Gemma Team, Aishwarya Kamath, Johan Ferret, et al. Gemma 3 technical report. *arXiv preprint arXiv:2503.19786*, 2025. doi: 10.48550/arxiv.2503.19786.
- Shahriar Golchin and Mihai Surdeanu. Time travel in LLMs: Tracing data contamination in large language models. *arXiv preprint arXiv:2308.08493*, 2023. doi: 10.48550/arxiv.2308.08493.
- Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, et al. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.
- Simeng Han, Hailey Schoelkopf, Yilun Zhao, et al. FO-LIO: Natural language reasoning with first-order logic. *arXiv preprint arXiv:2209.00840*, 2022. doi: 10.48550/arxiv.2209.00840.
- Ari Holtzman, Peter West, Vered Shwartz, Yejin Choi, and Luke Zettlemoyer. Surface form competition: Why the highest probability answer isn’t always right. *arXiv preprint arXiv:2104.08315*, 2021. doi: 10.48550/arxiv.2104.08315.
- Saurav Kadavath, Tom Conerly, Amanda Askell, et al. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022. doi: 10.48550/arxiv.2207.05221.
- Tamera Lanham, Anna Chen, Ansh Radhakrishnan, et al. Measuring faithfulness in chain-of-thought reasoning. *arXiv preprint arXiv:2307.13702*, 2023. doi: 10.48550/arxiv.2307.13702.
- Patrick Lewis, Ethan Perez, Aleksandra Piktus, et al. Retrieval-augmented generation for knowledge-intensive NLP tasks. *Advances in Neural Information Processing Systems*, 2020.
- Qing Lyu, Shreya Havaldar, Adam Stein, et al. Faithful chain-of-thought reasoning. *arXiv preprint arXiv:2301.13379*, 2023. doi: 10.48550/arxiv.2301.13379.
- Neil A. Macmillan and C. Douglas Creelman. *Detection Theory: A User’s Guide*. Lawrence Erlbaum Associates, 1991.
- Quinn McNemar. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 1947. doi: 10.1007/bf02295996.
- Microsoft, Abdelrahman Abouelenin, Atabak Ashfaq, et al. Phi-4-mini technical report: Compact yet powerful multimodal language models via mixture-of-loras. *arXiv preprint arXiv:2503.01743*, 2025. doi: 10.48550/arxiv.2503.01743.
- Theo Olausson, Alex Gu, Benjamin Lipkin, et al. LINC: A neurosymbolic approach for logical reasoning by combining language models with first-order logic provers. *arXiv preprint arXiv:2310.15164*, 2023. doi: 10.48550/arxiv.2310.15164.
- OLMo Team, Pete Walsh, Luca Soldaini, et al. 2 OLMo 2 furious. *arXiv preprint arXiv:2501.00656*, 2024. doi: 10.48550/arxiv.2501.00656.
- Liangming Pan, Alon Albalak, Xinyi Wang, and William Yang Wang. Logic-LM: Empowering large language models with symbolic solvers for faithful logical reasoning. *arXiv preprint arXiv:2305.12295*, 2023. doi: 10.48550/arxiv.2305.12295.
- Pouya Pezeshkpour and Estevam Hruschka. Large language models sensitivity to the order of options in multiple-choice questions. *arXiv preprint arXiv:2308.11483*, 2023. doi: 10.48550/arxiv.2308.11483.
- Qwen Team, Yang An, Baosong Yang, et al. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024. doi: 10.48550/arxiv.2412.15115.
- Abulhair Saparov and He He. Language models are greedy reasoners: A systematic formal analysis of chain-of-thought. *arXiv preprint arXiv:2210.01240*, 2022. doi: 10.48550/arxiv.2210.01240.
- Timo Schick, Jane Dwivedi-Yu, Roberto Dessì, et al. Toolformer: Language models can teach themselves to use tools. *arXiv preprint arXiv:2302.04761*, 2023. doi: 10.48550/arxiv.2302.04761.
- Freda Shi, Xinyun Chen, Kanishka Misra, et al. Large language models can be easily distracted by irrelevant context. *arXiv preprint arXiv:2302.00093*, 2023. doi: 10.48550/arxiv.2302.00093.
- Harold Stanislaw and Natasha Todorov. Calculation of signal detection theory measures. *Behavior Research Methods, Instruments, & Computers*, 1999. doi: 10.3758/bf03207704.
- Oyvind Tafjord, Bhavana Dalvi Mishra, and Peter Clark. ProofWriter: Generating implications, proofs, and abductive statements over natural language. *arXiv preprint arXiv:2012.13048*, 2020. doi: 10.48550/arxiv.2012.13048.
- Miles Turpin, Julian Michael, Ethan Perez, and Samuel R. Bowman. Language models don’t always say what they think: Unfaithful explanations in chain-of-thought

- prompting. *arXiv preprint arXiv:2305.04388*, 2023. doi: 10.48550/arxiv.2305.04388.
- Jason Wei, Xuezhi Wang, Dale Schuurmans, et al. Chain-of-thought prompting elicits reasoning in large language models. *Advances in Neural Information Processing Systems*, 2022. doi: 10.52202/068431-1800.
- Zhaofeng Wu, Linlu Qiu, Alexis Ross, et al. Reasoning or reciting? exploring the capabilities and limitations of language models through counterfactual tasks. *arXiv preprint arXiv:2307.02477*, 2023. doi: 10.48550/arxiv.2307.02477.
- Jian Xie, Kai Zhang, Jiangjie Chen, Renze Lou, and Yu Su. Adaptive chameleon or stubborn sloth: Revealing the behavior of large language models in knowledge conflicts. *arXiv preprint arXiv:2305.13300*, 2023. doi: 10.48550/arxiv.2305.13300.
- Kaiyu Yang, Aidan Swope, Alex Gu, et al. LeanDojo: Theorem proving with retrieval-augmented language models. *arXiv preprint arXiv:2306.15626*, 2023. doi: 10.48550/arxiv.2306.15626.
- Xi Ye, Qiaochu Chen, Işıl Dillig, and Greg Durrett. SatLM: Satisfiability-aided language models using declarative prompting. *arXiv preprint arXiv:2305.09656*, 2023. doi: 10.48550/arxiv.2305.09656.
- Zhangyue Yin, Qiushi Sun, Qipeng Guo, Jiawen Wu, Xipeng Qiu, and Xuanjing Huang. Do large language models know what they don’t know? *arXiv preprint arXiv:2305.18153*, 2023. doi: 10.48550/arxiv.2305.18153.
- Tony Z. Zhao, Eric Wallace, Shi Feng, Dan Klein, and Sameer Singh. Calibrate before use: Improving few-shot performance of language models. *arXiv preprint arXiv:2102.09690*, 2021. doi: 10.48550/arxiv.2102.09690.
- Chujie Zheng, Hao Zhou, Fandong Meng, Jie Zhou, and Minlie Huang. Large language models are not robust multiple choice selectors. *arXiv preprint arXiv:2309.03882*, 2023. doi: 10.48550/arxiv.2309.03882.